

Haralick Texture Features

Notation

N $p(i, j)$ μ_x μ_y σ_x σ_y k μ_{x+y} μ_{x-y} $p_{x+y}(k)$ $p_{x-y}(k)$ HX HY HXY $HXY1$ $HXY2$

1 Autocorrelation

[1], [2], [3]

$$\sum_{i=1}^N \sum_{j=1}^N (i \cdot j) p(i, j)$$

Put simply, autocorrelation gives the sum of the products of expected vlaues ($E[p(i, j)] = E[p_x(i)]E[p_y(j)] = (i \cdot j)p(i, j)$). If we assume N to be held constant, the distribution of the GLCM entries heavily determine the value of this feature. So if an image is relatively brighter autocorrelation will return a higher value.

As i or j increase the function increases. In terms of GLCMs as the indicies distance from the (1,1) entry increases the higher the wieghting of that entry; This means dark pixels co-occur with dark pixels, or low-intensity regions are dominant. The most bottom right entry (N, N) , has the highest possible weight. This means bright pixels co-occur with bright pixels. Also The top right and bottom left corners both have higher weights. In the image bright pixels co-occur with bright pixels.

2 Cluster Prominence

[2], [3]

$$\sum_{i=1}^N \sum_{j=1}^N (i + j - \mu_x - \mu_y)^4 p(i, j)$$

in [1] There is a typo, cluster prominence is raised to the third power, instead it should be raised to the fourth as per other sources. Also, $-\mu_x - \mu_y$ is written as -2μ where $\mu = (\mu_x + \mu_y)/2$.

μ_x and μ_y are the mean gray levels of the rows and columns of the GLCM. They represent the "center of mass" or the average intensity co-occurrence in the matrix.

The wieght can be rewritten as $(i - \mu_x) + (j - \mu_y)$, the sum of deviations from the mean. This term measures the distance of a specific co-occurrence (i, j) from the mean. The sum $(i + j)$ is compared against the sum of the means $(\mu_x + \mu_y)$, so this feature focuses on how much the texture "protrudes" from the average. Since the difference is raised to the 4th power, it heavily penalizes entries that are far away from the center of the GLCM. This would also ensure a positive value $((-a)^4 = a^4)$

Cluster prominence is a measure of the asymmetry of the image. A High cluster prominence indicates the image is not visually "smooth"; it suggests the presence of distinct, high contrast clusters (very bright or very dark areas) that stand out against the background (2μ). A low cluster prominence indicates that the GLCM is concentrated near the mean. The image is likely more uniform.

3 Cluster Shade

[2], [3]

$$\sum_{i=1}^N \sum_{j=1}^N (i + j - \mu_x - \mu_y)^3 p(i, j)$$

Typo in [1], mismatched with cluster prominence.

Similar to the interpretation of cluster prominence; except that this function is raised to the third power. This allows for negative values.

4 Cluster Tendency

[2], [3]

$$\sum_{i=1}^N \sum_{j=1}^N (i + j - \mu_x - \mu_y)^2 p(i, j)$$

Similar to the interpretation of cluster prominence.

4.1 Sum Variance

[1], [2], [3] (Mathematically equivalent to cluster tendency)

$$\sum_{k=2}^{2N} (k - \mu_{x+y})^2 p_{x+y}(k)$$

5 Contrast

[2], [3]

$$\sum_{i=1}^N \sum_{j=1}^N (i - j)^2 p(i, j)$$

The term $(i - j)^2$ acts as a weighting kernel that measures the squared distance from the main diagonal of the GLCM. When $i - j = 0$, the weighting is zero. These entries represent pixels with identical intensities occurring in the specified spatial relationship. The weighting increases quadratically as the distance $|i - j|$ increases. Consequently, this feature is highly sensitive to entries in the "far corners" of the matrix which represent transitions between extreme dark and extreme bright pixels.

If the probability mass of $p(i, j)$ is concentrated on or near the main diagonal, the summation remains small. In an image this translates to a homogeneous image where neighboring pixels have very similar gray levels. In this case, the GLCM is "diagonal-dominant".

If the mass is widely distributed across the off-diagonals, the contrast value grows. This indicates more frequent large intensity transitions. Which, in an image, can corresponds to sharp edges, high-frequency noise, or heavy texture.

6 Correlation

[2], [3]

$$\frac{\sum_{i=1}^N \sum_{j=1}^N (i \cdot j) p(i, j) - \mu_x \mu_y}{\sigma_x(i) \sigma_y(j)}$$

in [1], its written as:

$$\sum_{i=1}^N \sum_{j=1}^N \left(\frac{i - \mu_x}{\sigma_x} \right) \left(\frac{j - \mu_y}{\sigma_y} \right) p(i, j)$$

This is similar to a discrete implementation of the Pearson correlation coefficient for the joint probability mass function $p(i, j)$. It measures how much the gray level of a pixel is linearly predictable from its neighbor.

Entries where both i and j are simultaneously or singly greater than or less than the mean are positive (thus greater value). These represent "consistent" regions (dark-dark or bright-bright). Entries where one pixel is bright and the other is dark relative to the mean are; these represent "transitional" or "inconsistent" regions.

Since the terms are standardized by σ_x and σ_y , this feature is bounded (with respect to the GLCM).

7 Difference Average

[2]

$$\sum_{k=0}^{N-1} k \cdot p_{x-y}(k)$$

7.1 Dissimilarity

[1], [2], [3] (mathematically equivalent to Difference Average)

$$\sum_{i=1}^N \sum_{j=1}^N |i - j| p(i, j)$$

8 Difference Entropy

[2]

$$\sum_{k=0}^{N-1} p_{x-y}(k) \log_2 (p_{x-y}(k) + \epsilon)$$

in [3],

$$\sum_{i=0}^{N-1} p_{x-y}(i) \log_2 (p_{x-y}(i))$$

in [1],

$$\sum_{k=0}^{N-1} p_{x-y}(k) \log (p_{x-y}(k))$$

This feature uses a marginalization of the GLCM along its off-diagonals. It measures the uniformity of the distribution of differences. If the probability mass is concentrated on just a few values of k , the entropy is low. If the mass is distributed across all possible differences, entropy is maximized.

Shannon Entropy applied to the distribution of gray-level differences ($p_{x-y}(k)$). Simply Difference Entropy quantifies the complexity and disorder of the local intensity changes rather than the intensity of the changes themselves.

At a high difference entropy value the image contains a vast variety of local transitions (some subtle, some sharp) with no dominant pattern. There is high uncertainty in predicting the difference between a pixel and its neighbor. At a low difference entropy value the image has order or structural simplicity. The distribution of differences is sparse.

9 Difference Variance

[1], [2]

$$\sum_{k=0}^{N-1} (k - \mu_{x-y})^2 p_{x-y}(k)$$

this feature measures the variance of the marginalized difference distribution ($p_{x-y}(k)$). The term $(k - \mu_{x-y})^2$ acts as a weighting function that penalizes deviations from the mean difference μ_{x-y} . If an image has a very consistent texture where the intensity change between neighbors is stable, $p_{x-y}(k)$ will be highly concentrated around its mean, resulting in a low variance.

Low difference variance suggests a "predictable" texture. Even if the image is high-contrast, if the transitions between light and dark pixels follow a regular, repetitive rule, the variance of those differences remains small.

High difference variance suggests a complex texture where the relationships between adjacent pixels are not uniform. It captures the degree of disorder.

10 Energy

[2], [3] Also known as 'Joint Energy' and 'Angular Second Moment'.

$$\sum_{i=1}^N \sum_{j=1}^N p(i, j)^2$$

Energy is a measure of the concentration of the probability distribution within the GLCM. This feature operates on the principle that the sum of squares of a set of probabilities is maximized when the "probability mass" is concentrated in a few entries and minimized when it is spread uniformly. If an image is dominated by a certain co-occurrence, whether $i = j$ (a singular gray level) or $i \neq j$ (a singular gradient), the normalized GLCM at entry (i, j) will in turn have a large probability. This concentration (energy) at a singular entry causes the feature to increase; concentrations at multiple entries can produce a high feature value as long as its probabilities are much larger than the rest.

11 Entropy

[2] Also known as 'Joint Entropy'.

$$-\sum_{i=1}^N \sum_{j=1}^N p(i, j) \log_2 (p(i, j) + \epsilon)$$

[3]

$$-\sum_{i=1}^N \sum_{j=1}^N p(i, j) \log_2 (p(i, j))$$

[1]

$$-\sum_{i=1}^N \sum_{j=1}^N p(i, j) \log (p(i, j))$$

Shannon entropy applied to the normalized glcm. Entropy is maximized when the distribution is uniform ($p(i, j) = 1/N^2$ for all i, j), meaning every possible gray-level transition occurs with equal frequency. Entropy is minimized when the image is perfectly homogeneous, and only one entry in the GLCM has a probability of 1.

In the context of a GLCM, entropy quantifies the structural unpredictability of gray level co-occurrences. A low entropy GLCM indicates that only a few gray level pairs occur with significant probability (typical of highly regular textures, large uniform regions, or consistent gradients). A high entropy GLCM reflects a broad spread of co-occurrence probabilities, corresponding to complex, irregular, or noisy textures.

Entropy is agnostic to where the mass lies in the matrix (diagonal vs. off-diagonal). This feature can play as a compliment of Energy and Homogeneity-like features.

12 Homogeneity 1

[1], [2], [3] Also known as 'Inverse Difference'

$$\sum_{i=1}^N \sum_{j=1}^N \frac{p(i, j)}{1 + |i - j|}$$

Compared to Inverse difference moment, the denominator grows linearly rather than quadratically, Inverse difference doesn't nullify off-diagonal probabilities as quickly. This results in a feature that maintains a higher value even in the presence of low-level noise.

12.1 Inverse Difference Normalized

[2], [3]

$$\sum_{i=1}^N \sum_{j=1}^N \frac{p(i, j)}{1 + \frac{|i-j|}{N}}$$

13 Homogeneity 2

[2], [3] Also known as 'Inverse Difference Moment'

$$\sum_{i=1}^N \sum_{j=1}^N \frac{p(i, j)}{1 + |i - j|^2}$$

In [1], it is titled Homogeneity.

This feature quantifies the degree of local similarity in an image. The denominator term $1 + (i - j)^2$, acts as a penalty that is minimized when $i = j$ (the diagonal entries). In a region where pixel co-occurences are nearly identical, $(i - j) \rightarrow 0$, the weights approach 1. In a GLCM the diagonals and near diagonals will have the most contribution to the sum. As the gradients $(i - j)$ increase, the denominator grows quadratically which reduces the contribution of those entries to the total sum.

13.1 Inverse Difference Moment Normalized

[2], [3]

$$\sum_{i=1}^N \sum_{j=1}^N \frac{p(i, j)}{1 + \frac{|i-j|^2}{N^2}}$$

14 Information measure of correlation 1

[2], [3]

$$\frac{HXY - HXY1}{\max\{HX, HY\}}$$

15 Information measure of correlation 2

[2], [3]

$$\sqrt{1 - e^{-2(HXY2 - HXY)}}$$

16 Inverse Variance

[2]

$$\sum_{k=1}^{N-1} \frac{p_{x-y}(k)}{k^2}$$

[3]

$$\sum_{i=1}^N \sum_{j=1}^N \frac{p(i, j)}{|i-j|^2}, \quad i \neq j$$

The $\frac{1}{|i-j|^2}$ is similar to Inverse difference moment. Instead this feature begins at $k = 1$ to avoid division by 0. Because k^2 grows quadratically, the weights $1/k^2$ decreases quickly as we move closer to the diagonal. For example, a difference of $k = 1$ has a weight of 1.0, while a difference of $k = 10$ has a weight of 0.01. This feature has stronger weights for off-diagonals further from the diagonal, but instead of factoring by k (as done in Contrast) it is factored by $1/k^2$.

High Inverse Variance suggests a homogeneous image where majority of the neighboring pixels have identical or nearly identical gray levels. (plots of Inverse variance and contrast may seem identical because only two gray levels are used in our experiments).

17 Joint Average

[2]

$$\sum_{i=1}^N \sum_{j=1}^N i \cdot p(i, j)$$

Use only on symmetric GLCMs.

Joint Average acts as a proxy for the local brightness level of the textured region. This feature is half of the Sum Average.

18 Maximal Correlation Coefficient

[1], [2]

$$\sqrt{\lambda_2 Q(i, j)}$$
$$Q(i, j) = \sum_{k=0}^N \frac{p(i, k)p(j, k)}{p_x(i)p_y(k)}$$

19 Maximum Probability

[2], [3]

$$\max (p(i, j))$$

This feature simply returns the largest probability of co-occurrences from a GLCM. This feature gives less information about the texture, instead it may indicate if an image is dominated by a certain co-occurrence.

20 Sum Average

[2]

$$\sum_{k=2}^{2N} k \cdot p_{x+y}(k)$$

[3]

$$\sum_{i=2}^{2N} i \cdot p_{x+y}(i)$$

This gives us the sum of the indices weighted by the k value. Sum Average is essentially calculating the center of mass along the axis perpendicular to the main diagonal (the $i + j$ lines). It considers pixels that actually participate in the co-occurrence relationship (in a cross diagonal there is a certain set of co-occurrences). Higher weight is given to brighter values.

e.g The cross-diagonal that intersects the $(3, 3)$ entry of the GLCM, $4 \leq k \leq 6$ The feature compares the average within each cross diagonal (instead of the average of the whole GLCM), this captures variances with all the co-occurrences where the gray level co-occurs with gray level = 3. But gray levels greater than 3 are not captured.

21 Sum Entropy

[2]

$$\sum_{k=2}^{2N} p_{x+y}(k) \log_2 (p_{x+y}(k) + \epsilon)$$

[3]

$$- \sum_{i=2}^{2N} p_{x+y}(i) \log_2 (p_{x+y}(i))$$

[1]

$$- \sum_{k=2}^{2N} p_{x+y}(k) \log (p_{x+y}(k))$$

Sum Entropy is then the Shannon Entropy of the p_{x+y} distribution. Similar to Entropy, if the distribution is concentrated the Sum Entropy value will be near zero, and grows as the distribution is more uniform.

This feature can be used to describe the spatial distribution of brightness in a texture. High value indicates a texture where many different combinations of total brightness exist across the image. This is typical of heterogeneous or "noisy" environments where light and dark areas are mixed in many different ways.

22 Sum of Squares

[2]

$$\sum_{i=1}^N \sum_{j=1}^N (i - \mu_x)^2 p(i, j)$$

[1], [3] Denoted as 'Variance':

$$\sum_{i=1}^N \sum_{j=1}^N (i - \mu)^2 p(i, j)$$

References

- [1] Löfstedt, Tommy, et al. "Gray-Level Invariant Haralick Texture Features." PLOS ONE, vol. 14, no. 2, 22 Feb. 2019, p. e0212110, <https://doi.org/10.1371/journal.pone.0212110>.
- [2] Pyradiomics. <http://github.com/radiomics/pyradiomics>
- [3] Parmar, Chintan, et al. "Robust Radiomics Feature Quantification Using Semiautomatic Volumetric Segmentation." PLoS One, vol. 9, no. 7, 2014, p. e102107, [pubmed.ncbi.nlm.nih.gov/25025374/](https://doi.org/10.1371/journal.pone.0102107), <https://doi.org/10.1371/journal.pone.0102107>.